

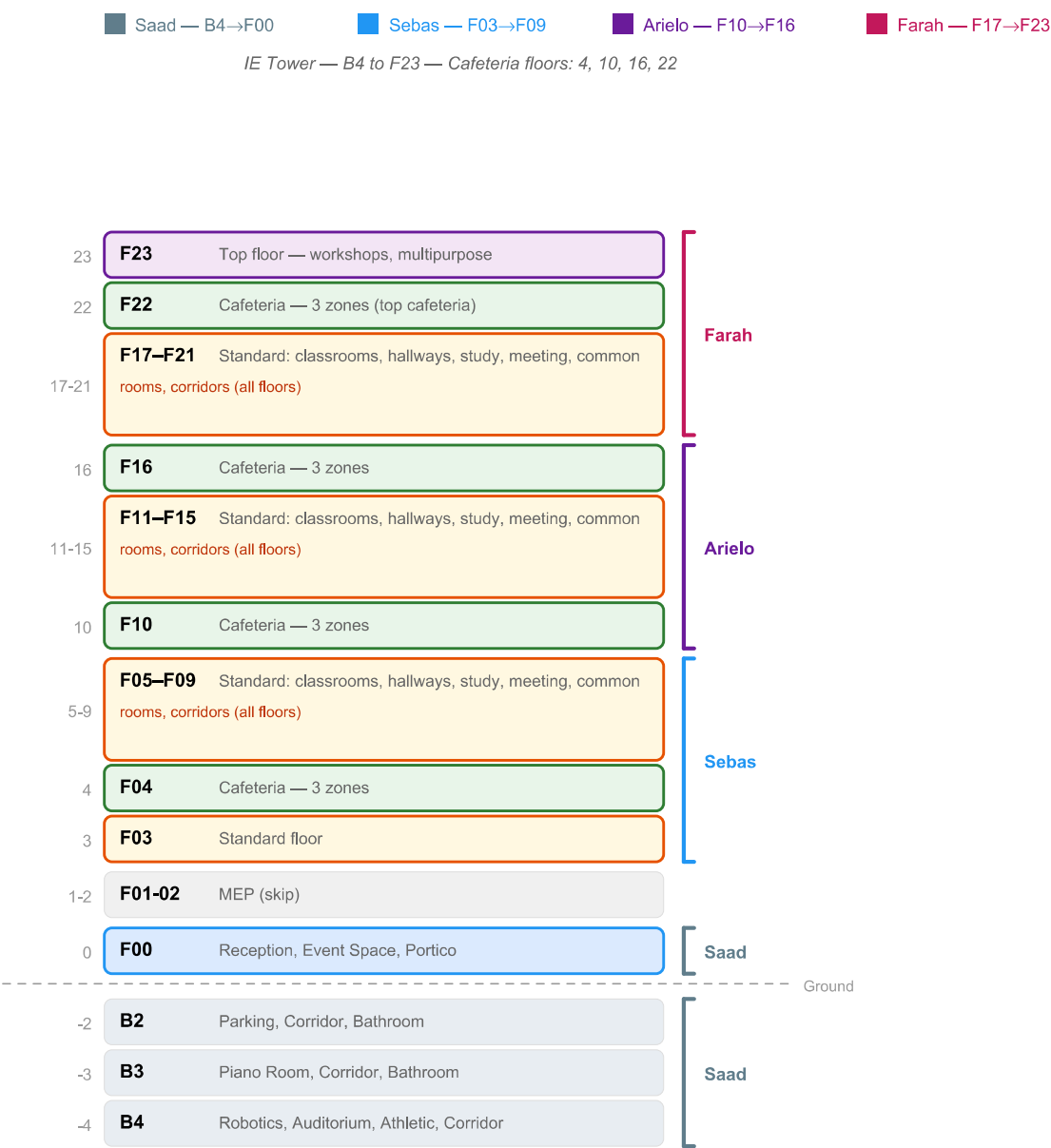
Location Labels & Workflow

IE Tower — VPR Project — Spring 2026

What to record, how to label it, and who does what

IE Tower — Cross Section

We cover B4 to F23. Green = cafeteria floors (4, 10, 16, 22). Yellow = standard classroom floors. Gray = basements. Purple = top floor.



Floor Plan Structure

- **Center:** main classroom (half, normal, or double depending on floor)
- **Left & right wings:** common rooms for students
- **Corridors:** connecting wings to elevators
- **Side rooms:** study rooms, meeting rooms along corridors
- **Cafeteria floors (4, 10, 16, 22):** central lobby + cafeteria side + opposite side — 3 locations each

1. Labeling Convention

Every clip and frame gets **one location label**. The label = **where**. Not when, not with what camera.

Format

floor_location_detail		
Part	Rule	Examples
floor	f## above ground, b# basement	f04, f22, b4, f00
location	Space type	cafeteria, classroom, hallway, study_room, meeting_room
detail	What makes it specific	half, normal, double, curved, north, east, main, seating

Good examples

f04_cafeteria_main f00_reception_turnstiles b4_robotics_lab f07_classroom_double
f12_hallway_north f05_study_room_east f22_cafeteria_seating f14_meeting_room_main
b4_auditorium_main

Don't do this

- ✗ Don't put lighting in the label (`_day`, `_night`) — metadata
- ✗ Don't put device in the label (`_blackmagic`, `_webcam`) — metadata
- ✗ Don't use generic labels like just 'hallway' — add a distinguishing detail
- ✗ Don't create different labels for the same place under different conditions

Rule: same place, same floor → same label. Different look or different floor → different label.

2. Metadata

Lighting and device go in the CSV, not the label:

```
image_path,label,split,device,lighting  
f04_cafeteria_main/frame_0001.jpg,f04_cafeteria_main,train,blackmagic,day  
f07_classroom_double/frame_0001.jpg,f07_classroom_double,train,blackmagic,night  
b4_auditorium_main/frame_0005.jpg,b4_auditorium_main,query,webcam,day
```

Split: ~80% → train, ~20% → query. Webcam footage always → query.

3. All Locations by Floor

Basements — B4, B3, B2 (Saad)

Label	What it is	Look for
b4_robotics_lab	Robotics / maker lab	Equipment, workbenches, robots
b4_auditorium_main	Auditorium seating	Dark seats, LED strips, stage
b4_auditorium_stage	Auditorium stage	Podium, screens, front view
b4_athletic_center	Athletic center / gym	Gym equipment, sports facilities
b4_corridor_main	Main corridor B4	Exposed pipes, signage
b4_bathroom_main	Bathroom B4	Different from tower bathrooms
b3_piano_room	Piano practice room	Piano, acoustic panels
b3_corridor_main	Corridor B3	Signage, doors
b3_bathroom_main	Bathroom B3	Distinctive fixtures
b2_parking_entrance	Parking entrance	Cars, ramps, signage
b2_corridor_main	Corridor B2	Concrete, signage
b2_bathroom_main	Bathroom B2	Only if visually different

Ground Floor — F00 (Saad)

Label	What it is	Look for
f00_reception_turnstiles	Main entrance + turnstiles	LED screen, gates, glass facade
f00_event_space	Event space / main hall	Open area, high ceiling
f00_portico_entrance	Portico / covered entrance	Concrete columns, exterior view

F01–F02 are MEP / mechanical — skip them.

Cafeteria Floors — F04, F10, F16, F22

3-zone structure. Each zone = separate location: **central lobby**, **cafeteria side**, **opposite side**.

Floor 04 (Sebas)

Label	What it is	Look for
f04_lobby_central	Central lobby	Elevators, unique furniture, ceiling
f04_cafeteria_main	Cafeteria area	Food counters, tables, kitchen view
f04_cafeteria_seating	Cafeteria seating	Tables, windows, lighting
f04_opposite_lounge	Opposite side	Sofas, study desks, different vibe

Floor 10 (Arielo)

Label	What it is	Look for
f10_lobby_central	Central lobby	Elevators, unique furniture, ceiling
f10_cafeteria_main	Cafeteria area	Food counters, tables, kitchen view
f10_cafeteria_seating	Cafeteria seating	Tables, windows, lighting
f10_opposite_lounge	Opposite side	Sofas, study desks, different vibe

Floor 16 (Arielo)

Label	What it is	Look for
f16_lobby_central	Central lobby	Elevators, unique furniture, ceiling
f16_cafeteria_main	Cafeteria area	Food counters, tables, kitchen view
f16_cafeteria_seating	Cafeteria seating	Tables, windows, lighting
f16_opposite_lounge	Opposite side	Sofas, study desks, different vibe

Floor 22 (Farah)

Label	What it is	Look for
f22_lobby_central	Central lobby	Elevators, unique furniture, ceiling
f22_cafeteria_main	Cafeteria area	Food counters, tables, kitchen view
f22_cafeteria_seating	Cafeteria seating	Tables, windows, lighting
f22_opposite_lounge	Opposite side	Sofas, study desks, different vibe

Standard Floors — F03, F05–F09, F11–F15, F17–F21 (ALL floors)

Cover every floor. Identify what's on each floor when you arrive.

Classroom types

Label pattern	What it is	Look for
<code>fXX_classroom_half</code>	Half-size classroom	Smaller room, ~30 seats
<code>fXX_classroom_normal</code>	Normal classroom	Standard size, rows, projector
<code>fXX_classroom_double</code>	Double classroom	Large room, possibly dividable
<code>fXX_classroom_curved</code>	Curved-desk classroom	Curved wooden desks, colored chairs

Other spaces

Label pattern	What it is	Look for
<code>fXX_common_room_east</code>	Common room (east)	Study tables, glass walls, carpet
<code>fXX_common_room_west</code>	Common room (west)	May differ in furniture
<code>fXX_study_room_[detail]</code>	Study room	Enclosed room, desks, quiet
<code>fXX_meeting_room_[detail]</code>	Meeting room	Glass walls, conference table, screen
<code>fXX_hallway_[detail]</code>	Hallway	Must have visual landmark
<code>fXX_corridor_elevator</code>	Elevator corridor	Elevator doors, floor signage

Replace `XX` with floor number (e.g. `f07_classroom_double`). For `[detail]` use compass direction, room number, or a descriptive word.

Top Floor — F23 (Farah)

Label	What it is	Look for
<code>f23_workshop_space</code>	Workshop / multipurpose	Open layout, large tables
<code>f23_corridor_main</code>	Main corridor	Signage, structural details
<code>f23_common_room_[detail]</code>	Common area	Identify when you get there

Bathrooms

Tower bathrooms look identical. Only record B2, B3, B4 bathrooms. Max 1-2 tower bathrooms if clearly distinct.

Max 5 bathroom locations total. More = near-identical clusters in embedding space.

4. Your Assignment — Per Person

Saad Ayomide — B4, B3, B2, F00

You have the **most unique spaces** — auditorium, robotics lab, athletic center, reception, piano room. None repeat. Easiest to label because everything looks different.

Basement lighting is tricky — artificial only, low ceilings. You'll need ISO closer to 400 and shutter at 1/60 down there. For day vs night: basements won't change much, but record at two different times anyway.

Example labels:

```
b4_robotics_lab    b4_auditorium_main    b4_auditorium_stage    b4_athletic_center    b4_corridor_main
b3_piano_room     b3_corridor_main     b2_parking_entrance    f00_reception_turnstiles    f00_event_space
f00_portico_entrance
```

Extra — Athletic Center: find it in the basement. Record the gym/sports area as a separate location. If it has multiple zones (weights, court), split them.

Sebas — F03, F04, F05, F06, F07, F08, F09

You have the **F04 cafeteria (3 zones)** plus 6 standard floors. F03 might have limited access — record whatever is accessible. For F05-F09, identify the classroom type on each floor and label accordingly.

F05-F09 will look similar. Differentiators: classroom type/size, window views (lower floors = closer buildings), floor number signage, furniture variations. **Always capture the floor number sign near the elevator.**

Example labels:

```
f03_classroom_normal    f03_corridor_elevator    f04_lobby_central    f04_cafeteria_main
f04_cafeteria_seating    f04_opposite_lounge    f05_classroom_half    f05_common_room_east
f05_hallway_north    f06_classroom_double    f07_classroom_curved    f07_meeting_room_main
f08_hallway_south    f09_study_room_east
```

Arielo Moreira — F10, F11, F12, F13, F14, F15, F16

You have **two cafeteria floors (F10 + F16)** — 8 cafeteria zone labels — plus 5 standard floors. Mid-tower floors have mixed natural/artificial light and window views start showing more sky.

F10 and F16 cafeterias look different from each other — different furniture, ceiling, view height. Make sure labels reflect the floor number. On standard floors, capture floor signage and window views.

Example labels:

```
f10_lobby_central    f10_cafeteria_main    f10_cafeteria_seating    f10_opposite_lounge
f11_classroom_normal    f11_hallway_east    f12_classroom_double    f13_meeting_room_main
f14_common_room_east    f15_corridor_elevator    f16_lobby_central    f16_cafeteria_main
f16_cafeteria_seating    f16_opposite_lounge
```

Extra — Laptop webcam pass: after your Blackmagic recordings, go back to 5-8 iconic spaces across the building (reception, a cafeteria, auditorium, a hallway) and record 10-15 sec with your laptop webcam. No special settings. Small dataset for robustness against lower-quality queries.

Farah Habib — F17, F18, F19, F20, F21, F22, F23

You have the **F22 cafeteria (top cafeteria, 3 zones)** + F23 (top floor, workshops) + 5 standard floors. Upper floors have the best window views — more sky, city panorama — naturally distinguishable from lower floors.

F22 is the highest cafeteria — views are very different from F04 or F10. Capture those windows. F23 is unique (workshops, multipurpose). Walk around and label whatever you find up there.

Example labels:

```
f17_classroom_half    f17_hallway_south    f18_classroom_double    f18_common_room_west
f19_study_room_east    f20_meeting_room_main    f21_corridor_elevator    f22_lobby_central
f22_cafeteria_main    f22_cafeteria_seating    f22_opposite_lounge    f23_workshop_space
f23_corridor_main
```

5. Recording & Processing Workflow

1. **Record clips** — one clip per location, per lighting condition. Blackmagic Camera (see Recording Guide).
2. **Name the clip:** `label__lighting__device.mov`
e.g. `f07_classroom_double__day__blackmagic.mov`
3. **Split long videos** if needed:
`ffmpeg -i long_video.mov -ss 00:01:30 -t 00:00:25 -c copy clip.mov`
4. **Extract frames** — 1 frame/sec:
`python -m src.data.extract_frames --video clip.mov --output data/processed_frames/LABEL/ --fps 1`
5. **Update CSV** — one row per frame in `data/metadata/dataset.csv`
6. **Push to repo** — `git add frames + CSV, commit, push.`

Folder structure

```
data/processed_frames/  
  f07_classroom_double/  
    frame_0001.jpg ← day  
    frame_0002.jpg  
    ...  
    frame_0021.jpg ← night  
  f07_hallway_north/  
    frame_0001.jpg  
    ...
```

6. Per-Location Checklist

#	Task	
1	Record Blackmagic clip — DAY lighting	
2	Record Blackmagic clip — NIGHT lighting	
3	Record webcam clip (Arielo — key spaces only)	
4	Split long videos into individual clips	
5	Rename clips with naming convention	
6	Extract frames (1 fps) into labeled folders	
7	Add rows to dataset.csv	
8	Double-check label spelling	
9	Push to GitHub	

Each person handles the full pipeline for their floors: record → split → label → extract → CSV → push.